

Paper Name: Research Methodology (Semester 6)

**The Impact of Generative AI
on Critical Thinking,
Problem-Solving Skills And
Personal Use Among
Undergraduate Students**

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Introduction

In recent years, the use of generative artificial intelligence (AI) has increased rapidly, especially in the field of education. Tools such as ChatGPT, Google Gemini, and Claude are now commonly used by undergraduate students for completing assignments, solving problems, and understanding academic concepts. These tools provide quick and structured responses, making academic work easier and faster.

However, this growing use of AI has raised an important concern: **Are students becoming too dependent on AI?** Traditionally, education focuses on developing important skills such as critical thinking and problem-solving. Critical thinking means the ability to analyse information, question it, and form independent judgments. Problem-solving refers to the ability to logically work through challenges and find solutions.

The issue arises when students start relying heavily on AI tools instead of thinking independently. For example, instead of attempting a question themselves, students may directly ask AI for answers. While this saves time, it may reduce their ability to think deeply or solve problems on their own.

At the same time, AI is not entirely harmful. It can also help students understand difficult topics, provide different perspectives, and improve learning efficiency. This creates a **dual situation**—AI can either support learning or weaken important cognitive skills depending on how it is used.

This topic is highly important in real life because AI tools are becoming a regular part of student life. If overused, they may negatively impact students' intellectual growth and future performance in jobs where independent thinking is required.

Therefore, this study aims to examine the impact of generative AI on critical thinking and problem-solving skills among undergraduate students. It focuses on understanding whether AI is helping students learn better or making them more dependent.

Literature Review

Study 1: Role of Usage Behaviour and Self-Regulation

The study by Atiyah A. Alghamdi (2025) explores the intersection of Generative AI (GenAI) and higher-order cognitive skills in university settings. As tools like ChatGPT become ubiquitous, educators are concerned with whether AI serves as a "cognitive scaffold"—a mental support structure that helps students reach higher levels of analysis—or if it leads to "cognitive passivity," where students offload deep reasoning to the machine. The research specifically aims to understand how a student's academic discipline (STEM vs. Non-STEM) and their AI self-efficacy (confidence in using the technology) shape their perceptions of AI's impact on their critical thinking and creativity.

Methodology

Alghamdi employed a quantitative, cross-sectional survey design, collecting data from 405 undergraduate students at a university in Saudi Arabia. The sample was composed of 245 male and 160 female students, with 73.6% enrolled in STEM fields and 26.4% in non-STEM fields. Participants responded to structured 5-point Likert scales adapted from established frameworks, such as the California Critical Thinking Disposition Inventory. The researcher used independent t-tests to identify differences between academic disciplines and simple linear regression to measure how strongly AI self-efficacy predicted a student's positive outlook on the technology.

Key Findings

The results highlighted a "moderately favorable" view of GenAI, with several nuanced distinctions:

- **Disciplinary Divergence:** A significant difference was found in critical thinking perceptions; STEM students were much more positive about AI's role in analytical reasoning than non-STEM students. This suggests that AI tools align more naturally with the logic-based, computational tasks prevalent in STEM curricula.
- **Creative Convergence:** Interestingly, there was no significant difference between disciplines regarding creativity. Students across all fields moderately agreed that AI helps with idea generation and "thinking outside the box."
- **The Critical Role of Self-Efficacy:** AI self-efficacy was a powerful predictor of perception. It accounted for 36.4% of the variance in critical thinking and a substantial 65.1% of the variance in creativity. Essentially, the more confident a student feels using AI, the more they view it as a tool for intellectual growth rather than a shortcut.

- **Awareness of Limitations:** While students valued AI for synthesis and evaluation, the scores remained moderate (averaging around 3.3 to 3.6), indicating a "cautious optimism" and an awareness of the risks of over-reliance.

Conclusion

Alghamdi concludes that GenAI is perceived as a valuable cognitive partner, but its effectiveness depends heavily on the learner's background and confidence. The study suggests that for AI to remain a scaffold rather than a crutch, higher education must move beyond basic technical training. Institutions should implement discipline-specific AI literacy that teaches students how to critically interrogate AI outputs. By fostering high AI self-efficacy and designing assessments that focus on the "process" of thinking rather than just the final product, educators can ensure that GenAI enhances, rather than replaces, human cognition.

Study 2: Impact on Critical Thinking Skills

The study conducted by Brazão and Tinoca, published in *Frontiers in Education* (2025), provides a comprehensive examination of the role of artificial intelligence, particularly AI-powered educational chatbots, in influencing students' critical thinking abilities. A key strength of this study lies in its structured analysis of how students engage cognitively with AI systems through different levels of questioning. The primary objective of the study was to investigate how interaction with AI chatbots shapes students' critical thinking processes, specifically aiming to determine whether AI promotes higher-order thinking or encourages passive learning behaviours.

Methodology

The researchers adopted a qualitative case study approach, analysing students' interactions with AI chatbots. A significant feature of the methodology was the classification of students into five distinct groups, based on their patterns of engagement and questioning while interacting with the AI system. Additionally, the study introduced a five-level framework of questioning, which was used to evaluate the depth of students' critical thinking.

Five Levels of Critical Thinking (Questioning Framework)

The study categorises students' thinking into the following progressive levels, which highlights the shift from basic understanding to advanced critical evaluation.

1. **General and Defining Questions:** Basic understanding and simple explanations
2. **Specific Questions:** More focused, topic-related inquiries

3. **Applied Questions:** Use of knowledge in real-life situations
4. **Integrative Questions:** Combining ideas and perspectives
5. **Critical Engagement Questions:** Evaluation and questioning of AI responses (highest level)

Grouping of Students

Students were divided into five groups based on their interaction patterns, enabling researchers to analyze differences in AI use across varying levels of cognitive engagement.

- Lower-level groups primarily asked general and defining questions
- Intermediate groups showed the ability to ask applied and integrative questions
- Higher-level groups engaged in critical questioning, demonstrating deeper analytical and evaluative thinking

Key Findings

The findings of the study reveal a **dual and context-dependent impact** of AI on critical thinking.

- **Positive:** AI can enhance understanding, promote inquiry-based learning, and support higher-order thinking when used actively. Under guided conditions, AI was found to support the development of higher-order thinking skills, including analysis, evaluation, and reflection.
- **Negative:** Over-reliance on AI can lead to passive learning, reduced critical evaluation, and weaker independent thinking. These findings suggest that unregulated use of AI can lead to a decline in students' ability to engage in deep cognitive processing.

Conclusion

The study concludes that the impact of AI on critical thinking is inherently **contextual and usage-dependent**. While AI chatbots possess the potential to enhance cognitive development, their effectiveness is contingent upon **active, reflective, and critical engagement** by students. The researchers emphasize the necessity for **structured pedagogical frameworks and guided AI integration** to ensure that such technologies complement rather than replace students' intellectual efforts.

Study 3: AI and Problem-Solving Ability

The study by Renato P. dos Santos addresses a very relatable problem—why students often struggle with Chemistry. It often feels abstract, confusing, and hard to connect with real life. To

tackle this, the study explores how AI chatbots like ChatGPT and Bing Chat can act as "agents-to-think-with" rather than just tools that give answers. Instead of spoon-feeding information, the goal is to see if these chatbots can help students think, question, and understand complex concepts through a conversation—almost like a friendly tutor who adjusts to your specific pace and confusion.

Methodology

The researcher used a comparative case study method to analyze how students interacted with two different AI systems: ChatGPT and Bing Chat. The study looked at extensive interaction logs (the "chat history") from simulated Chemistry learning experiences. By using a "constructionist" lens—a theory that says we learn best by making and doing things—the researcher examined how well each bot could handle specific Chemistry problems, explain molecular structures, and react to different "prompts" or instructions given by the students.

Key Findings

The study showed that AI has the potential to make subjects like Chemistry feel much less intimidating:

- **Interactive Tutors:** AI chatbots made learning more personalized by using simple examples and analogies to explain "invisible" chemical processes.
- **The Socratic Approach:** When programmed correctly, the bots used a question-based method that forced students to think for themselves rather than just copying an answer.
- **The Battle of the Bots:** ChatGPT performed better than Bing Chat; its responses were more detailed, clear, and context-aware, whereas Bing Chat was often shorter and less engaging.
- **The Art of the Prompt:** A major takeaway was that the quality of learning depends on the student. Better prompting = better explanations. Learning became a two-way process rather than passive memorization.

Conclusion

Dos Santos concludes that AI chatbots can revolutionize STEM education by turning a one-way lecture into a back-and-forth conversation. However, the paper gives a necessary reality check: AI can still make mistakes ("hallucinate"), so critical thinking is still the most important tool a student has. The study suggests that with the right educator training and a focus on "prompt crafting," AI can turn a difficult subject like Chemistry into something engaging, inclusive, and much easier to understand.

Research Gap

- The "Perception vs. Performance" Divide (Empirical Evidence) The study by Alghamdi (2025) relies entirely on self-reported surveys, measuring how students feel AI helps them. Similarly, Sichterman et al. (2025) focus on qualitative peer feedback. There is a critical lack of empirical data that measures actual performance in critical thinking. This pilot study can bridge this by looking for measurable shifts in how students solve problems, rather than just asking them if they like the tool.
- Missing Link Between Critical Thinking and Problem-Solving While dos Santos (2023) looks at problem-solving in Chemistry and Alghamdi (2025) looks at critical thinking and creativity, these variables are rarely analyzed together in a single framework. Since these two skills are two sides of the same coin in undergraduate education, this research fills a gap by examining how AI influences the interaction between thinking critically and arriving at a solution.
- Disciplinary Silos (Beyond STEM and Chemistry) Current research is often "siloed." For example, dos Santos focuses strictly on Chemistry, and Alghamdi notes a massive gap in how non-STEM students perceive AI. There is a lack of research that looks at the general undergraduate population to see if AI's impact on problem-solving is a universal cognitive change or if it remains restricted to specific technical subjects.
- The Usage Pattern "Black Box" While Sichterman et al. (2025) discuss "over-reliance," none of the studies statistically break down usage patterns such as how often a student uses AI (frequency) versus the complexity of the problems they solve. Most studies treat AI use as a "yes/no" variable. Our study has the opportunity to analyze if different levels of dependency lead to different cognitive outcomes.
- Lack of Small-Scale "Pilot" Contexts Most existing literature consists of either very broad theoretical papers or large-scale data analyses. There is a significant shortage of small-scale quantitative pilot studies that allow for a closer look at the undergraduate transition. A pilot study like this is necessary to establish the "baseline" behaviors of students before large-scale institutional policies are implemented.
- Practical "How-To" Guidance for Educators While all three papers mention that AI can be a cognitive scaffold, they provide very few data-driven strategies for how a student can avoid cognitive passivity. There is a gap in research that provides actionable results backed by correlation and regression to show exactly at what point AI stops being a helper and starts being a crutch.

Objective

The main objectives of this study are:

- To examine the relationship between generative AI usage and critical thinking skills
- To analyze the impact of AI on problem-solving ability
- To determine whether high AI usage leads to dependency among students

Data Collection

- Type of Data: Primary data
- Method: Questionnaire
- Sample: Undergraduate students
- Sampling Method: Convenience sampling

Statistical Methods Used

The following statistical techniques will be applied:

- Descriptive Statistics (Mean, Standard Deviation)
- Correlation Analysis (to study relationships)
- Regression Analysis (to measure impact)
- Hypothesis Testing (t-test) (to compare groups)

Plan of Work

- Design and distribute questionnaire
- Collect responses from students
- Code and organize data in Excel
- Perform statistical analysis
- Interpret results and draw conclusions

Redefining The Research Objectives

1. To understand how the use of generative AI affects the way undergraduate students think critically.
2. To explore whether AI tools help students improve their problem-solving skills or make them rely too much on ready-made answers.
3. To examine if students who use AI more frequently tend to become dependent on it for completing their academic work.
4. To find out whether AI acts as a helpful learning support or reduces students' ability to think independently.
5. To compare how students with different levels of AI usage differ in their thinking and problem-solving abilities.
6. To study how different ways of using AI influence the balance between learning independently and taking help from technology.

Hypothesis Formulation

1. Relationship between AI usage and critical thinking

- H_0 (Null Hypothesis): There is no significant relationship between the use of generative AI and students' critical thinking skills.
- H_1 (Alternative Hypothesis): There is a significant relationship between the use of generative AI and students' critical thinking skills.

2. Impact on problem-solving ability

- H_0 : Generative AI has no significant impact on students' problem-solving skills.
- H_1 : Generative AI has a significant impact on students' problem-solving skills.

3. Impact of AI using personal use

- H_0 : Generative AI has no significant impact on students' personal life
- H_1 : Generative AI has a significant impact on students' personal life.

Type of Data

This study is based on primary data. Primary data refers to the data that is collected first-hand by the researcher for a specific purpose.

In this study, the data has been collected directly from undergraduate students through a structured questionnaire. The data is original in nature and has not been previously used in any other research.

The source of data is the responses obtained from students of different courses and academic years through an online survey conducted using Google Forms.

The Sampling Design

Population

The population of the study consists of undergraduate students enrolled in colleges. The study focuses on students who are likely to use generative AI tools for academic purposes.

Sample Size

A sample of approximately 30–50 undergraduate students has been selected for the study.

This sample size is considered appropriate for a pilot study, as it provides preliminary insights while remaining manageable within limited time and resources.

Sampling Technique

The study uses convenience sampling, which is a non-probability sampling method.

Under this method, participants are selected based on their availability and willingness to respond to the questionnaire.

Convenience sampling has been chosen due to the following reasons:

- The study is a pilot study, which does not require a very large or highly representative sample
- Limited time and resources restrict the use of more complex sampling methods
- Easy access to respondents (college students) ensures quick and efficient data collection
- It is widely used in exploratory research to obtain initial insights

Data Collection Method

The data for the study has been collected using a structured questionnaire.

The questionnaire was designed to measure:

- Usage of generative AI tools
- Critical thinking ability
- Problem-solving skills

The questionnaire consists of close-ended questions based on a Likert scale (1 to 5), where respondents indicate their level of agreement with each statement.

The survey was distributed online through Google Forms, and responses were collected electronically.

Mode of Data Collection

- Online survey method
- Shared via digital platforms such as WhatsApp and other communication channels

Advantages of This Method

- Easy and quick data collection
- Cost-effective
- Allows participation from multiple respondents
- Ensures uniformity in responses

Variables Identification

Independent Variable

- Generative AI Usage
 - Measured through responses related to frequency, dependency, and purpose of AI use

Dependent Variables

- Critical Thinking Skills
 - Measured using responses to analytical and evaluation-based questions
- Problem-Solving Ability
 - Measured through responses related to independent problem-solving and efficiency

Statistical Tools & Techniques

The following statistical methods will be used:

1. Descriptive Statistics

- Mean
- Standard Deviation

Used to summarize the overall trend of data

2. Correlation Analysis

- To measure the relationship between:
 - AI usage and critical thinking
 - AI usage and problem-solving

Helps in understanding the direction and strength of relationship

3. Regression Analysis

- To study the **impact of AI usage on critical thinking**
- Helps in identifying whether AI usage significantly affects cognitive skills

4. Hypothesis Testing (t-test)

- To compare:
 - High AI users vs Low AI users

Helps in testing whether differences between groups are statistically significant

Justification of Methods

These methods are suitable because they:

- Provide both **descriptive and inferential insights**
- Help in analyzing **relationships, impact, and differences**
- Are appropriate for **quantitative questionnaire data**

Scope and Limitations

Scope of the Study

- Focuses on undergraduate students
- Examines the impact of generative AI on cognitive skills
- Provides insights into modern learning behavior in the AI era

Limitations of the Study

- Small sample size (pilot study)
- Use of convenience sampling may cause bias
- Data is based on self-reported responses
- Limited time and resources
- Cannot establish causation, only relationships

DATA COLLECTION

The following questions were asked to collect the data using google form.

<https://docs.google.com/forms/d/1nIpS-UOy8MFZyovU9i7LSvXQYSsf754hgVgGGwkVYvo/edit>

DATA PRESENTATION AND DESCRIPTIVE ANALYSIS

In the following attached excel sheet the data is organised in a clear and structured format in table form for frequency distribution along with visual representation using graphs and pie charts.

<https://excel.cloud.microsoft/open/onedrive/?docId=7F88672BD1ED4F02%21s930b70ebab654dd9829e50980c32d54d&driveId=7F88672BD1ED4F02>

AI Elements	Average	median	mode	standard deviation	variance
AI usage	3.965686275	4	4	0.782655109	0.61254902
Problem solving	3.544117647	3.5	3.5	0.55835894	0.311764706
Critical thinking	3.289215686	3.25	3	0.691687337	0.478431373
Personal use	3.593137255	4	4	1.004938785	1.009901961

Interpretation of Descriptive Statistics

- The mean values between (3–4) indicate that most respondents agree that they use AI frequently
- Median and mode being close to 4 shows majority responses are “Agree” or “Strongly Agree”
- Personal use has highest mean, indicating strongest usage area
- Standard deviation is relatively low (0.5–1), meaning: Responses are not highly spread out as most students have similar opinions

- In simple terms: Students consistently report moderate to high usage of AI, especially for personal purposes.

Application of Statistical Techniques and their Results and Interpretation

Correlation Analysis

Formula

$$r = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2 \cdot \sum (Y_i - \bar{Y})^2}}$$

Results

- AI & Personal Use $\rightarrow r \approx 0.51$ (Moderate positive)
- AI & Problem Solving $\rightarrow r \approx 0.18$ (Weak positive)
- AI & Critical Thinking $\rightarrow r \approx -0.11$ (Weak negative)

Regression Analysis

Model

$$Y = a + bX$$

Steps

1. Selected AI usage as X
2. Selected dependent variables (Y):
 - Personal Use
 - Critical Thinking
 - Problem Solving
3. Used Excel **Data Analysis** → **Regression**
4. Obtained:
 - R^2
 - Coefficients
 - p-values

Variable	R^2	p-value	Conclusion
Personal Use	0.327	<0.05	Significant
Critical Thinking	0.013	>0.05	Not significant
Problem Solving	0.026	>0.05	Not significant

Decision Rule

$p < 0.05 \rightarrow \text{Reject } H_0$

$p > 0.05 \rightarrow \text{Accept } H_0$

Final Decisions

- Personal Use → Reject H_0 (Significant)
- Critical Thinking → Accept H_0 (Not significant)
- Problem Solving → Accept H_0 (Not significant)

Overall Interpretation

- AI is widely used and accepted by students

- It has a clear impact on personal use
- However, it does not significantly improve higher-level skills
- **Final Insight:** AI is being used more for convenience rather than for developing critical thinking or problem-solving abilities.

Findings of the Study

- A majority of respondents reported frequent use of AI tools in their daily academic and personal activities.
- The descriptive statistics (mean values around 3–4) indicate that students generally agree or strongly agree that they use AI regularly.
- AI usage shows a moderate positive relationship with personal use, meaning that higher AI usage leads to greater reliance on AI for everyday tasks.
- The regression results confirm a statistically significant impact of AI on personal use ($p\text{-value} < 0.05$).
- There is a very weak and statistically insignificant relationship between AI usage and critical thinking ability.
- The correlation between AI usage and critical thinking is slightly negative, suggesting that excessive reliance on AI may reduce independent thinking.
- AI usage has a weak and insignificant relationship with problem-solving skills, indicating no meaningful improvement in this area.
- The hypothesis testing results show that:
 - Null hypothesis is rejected for personal use
 - Null hypothesis is accepted for critical thinking and problem-solving
- The standard deviation values indicate that responses are fairly consistent, with limited variation among participants.
- Overall, students tend to use AI primarily for convenience, efficiency, and quick solutions rather than for enhancing deeper cognitive skills.

Limitations of the Study

The study, being a pilot study, had certain limitations that may affect the generalization and accuracy of the results. These are as follows:

1. Data Limitations

- The data collected was primary and self-reported, which may include personal bias or inaccurate responses from participants.
- Respondents might have answered based on perception rather than actual behavior, affecting the reliability of the data.
- Only a limited number of variables were considered, and other important factors such as academic performance, creativity, and long-term learning outcomes were not included.

2. Sample Size

- The study was conducted on a small sample size (pilot study), which limits the ability to generalize the findings to a larger population.
- The sample may not fully represent all types of students, backgrounds, or institutions.
- A larger and more diverse sample would provide more accurate and reliable results

3. Time Constraints

- The study was conducted within a limited time period, which restricted the depth of analysis.
- Due to time limitations, advanced statistical techniques and broader data collection could not be applied.
- A longer study duration could have allowed for more detailed insights and better validation of results.

Scope for Future Research

The present study provides initial insights into the impact of AI usage on students; however, it can be further extended and improved in several ways:

Increase in Sample Size

- Future studies can include a larger and more diverse sample to improve the reliability and generalizability of the results.

- Including students from different colleges, courses, and age groups would provide a broader perspective.

Inclusion of Additional Variables

- The study can be expanded by incorporating other important factors such as:
 - Academic performance
 - Creativity
 - Decision-making ability
 - Learning outcomes
- This would give a more comprehensive understanding of AI's impact.

Use of Advanced Statistical Techniques

- Future research can apply more advanced methods such as:
 - Multiple regression analysis
 - Factor analysis
 - ANOVA
- This would allow for a deeper and more accurate analysis of relationships between variables.

Comparative Studies

- Future research can compare:
 - Students who use AI frequently vs those who do not
 - Different educational institutions or regions
- This would help identify patterns and differences in AI usage and its impact.

Practical Application-Based Research

- Studies can focus on how AI is used in real academic tasks rather than just survey responses.
- This would reduce bias and provide more realistic results.

Summary of the Study

Objective of the Study

The primary objective of this study was to examine and analyse the impact of Artificial Intelligence (AI) usage on students in different aspects of their academic and personal development. The study specifically focused on three key dimensions:

- **Personal Use:** To understand how frequently and effectively students use AI tools in their daily lives for tasks such as completing assignments, searching for information, and improving efficiency.
- **Critical Thinking:** To evaluate whether the use of AI enhances students' ability to think analytically, interpret information independently, and make logical decisions.
- **Problem-Solving Ability:** To assess whether AI tools help students develop better problem-solving skills or if they reduce the need for independent effort.

The study aimed to determine whether AI acts as a learning enhancer that improves cognitive abilities or primarily serves as a convenience tool that simplifies tasks without contributing to deeper understanding.

What the Analysis Revealed

The statistical analysis, including descriptive statistics, correlation, regression, and hypothesis testing, provided the following insights:

- **Significant Impact on Personal Use:** The results indicated a moderate and statistically significant relationship between AI usage and personal use. This means that students who use AI more frequently tend to rely on it heavily for everyday tasks. AI tools are being widely adopted for convenience, saving time, and improving efficiency in routine academic and non-academic activities.

- **No Significant Impact on Critical Thinking:** The findings showed a very weak and statistically insignificant relationship between AI usage and critical thinking. This suggests that the use of AI does not contribute to improving students' analytical or independent thinking abilities. In some cases, over-reliance on AI may reduce the effort required to think deeply, leading to a decline in critical engagement.
- **No Significant Impact on Problem-Solving Ability** similarly, the analysis revealed that AI usage does not have a meaningful effect on problem-solving skills. Students may be using AI to directly obtain answers rather than engaging in the process of solving problems themselves. As a result, their ability to develop logical reasoning and independent problem-solving remains unaffected.

Conclusion

Based on the findings, it can be concluded that AI has become an important part of students' daily lives, particularly as a tool for convenience and efficiency. However, its role in enhancing higher-order cognitive skills such as critical thinking and problem-solving is limited.

The study highlights a growing tendency among students to use AI as a shortcut tool, which may reduce active learning and independent effort. While AI offers several benefits in terms of speed and accessibility, it does not automatically lead to skill development unless used thoughtfully.

Thus, AI should be viewed as a supportive tool rather than a substitute for learning, and its use must be balanced to ensure that it contributes positively to students' overall intellectual growth.

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